Lecture 18

Type A: from Schur modules to highest weights

Objectives. *The nonzero Schur modules of the finite-group course are irreducible $\mathfrak{gl}_n$-modules. Their highest weights are partitions. They give every simple $\mathfrak{sl}_n$-module. The Schur bialternant identity is the Weyl character formula.*

By the end of this lecture you will be able to:

- Prove that $\mathbb{S}_\lambda(\mathbb{C}^n)$ with $\ell(\lambda) \le n$ is irreducible under $\mathfrak{gl}_n$ and under $\mathfrak{sl}_n$.
- Pass between partitions, Dynkin labels and highest weights.
- Explain determinant twists, full columns and the length cutoff.
- Realise a Schur module as the image of a Young symmetriser.

Throughout, $n \ge 2$. A partition $\lambda \vdash d$ of length $\ell(\lambda) \le n$ is padded with zeros to $(\lambda_1, \dots, \lambda_n)$. Write V^λ for the irreducible S_d-module of the finite-group course. All modules are finite-dimensional. The empty partition gives $(\mathbb{C}^n)^{\otimes 0} = \mathbb{C}$ and $\mathbb{S}_\varnothing(\mathbb{C}^n) = \mathbb{C}$, the trivial module. Every result below is immediate for this module, so the arguments take $d \ge 1$.

18.1 Schur–Weyl duality recalled

Let $d \ge 1$ and $T^d = (\mathbb{C}^n)^{\otimes d}$, with basis vectors

$$e_I = e_{i_1} \otimes \cdots \otimes e_{i_d}.$$

The symmetric group S_d permutes tensor factors:

$$\sigma \cdot (v_1 \otimes \cdots \otimes v_d) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(d)}.$$

The Lie algebra $\mathfrak{gl}_n$ acts by the tensor-product rule of Lecture 14:

$$\rho_d(X) = \sum_{k=1}^{d} X_{(k)}, \qquad X_{(k)} = \mathrm{id} \otimes \cdots \otimes X \otimes \cdots \otimes \mathrm{id} \quad (X \text{ in factor } k).$$

The subscript distinguishes ρ_d from the Weyl vector ρ. Conjugation by σ moves factor k to position $\sigma(k)$, so $\sigma X_{(k)} \sigma^{-1} = X_{(\sigma(k))}$. Summing over k shows that $\rho_d(X)$ commutes with every $\sigma \in S_d$.

Definition 18.1 (Schur module). For $\lambda \vdash d$, the *Schur module* is

$$\mathbb{S}_\lambda(\mathbb{C}^n) = \mathrm{Hom}_{S_d}(V^\lambda, T^d), \qquad X \cdot \varphi = \rho_d(X) \circ \varphi.$$

This is a $\mathfrak{gl}_n$-module because $\rho_d(X)$ commutes with S_d.

Two families are already familiar. For the one-row partition (d), the module $V^{(d)}$ is trivial. So $\mathbb{S}_{(d)}(\mathbb{C}^n)$ is the space of symmetric tensors, and averaging over S_d identifies it with $\operatorname{Sym}^d \mathbb{C}^n$. For the one-column partition (1^d), the module $V^{(1^d)}$ is the sign module. So $\mathbb{S}_{(1^d)}(\mathbb{C}^n)$ is the space of alternating tensors, and the antisymmetriser identifies it with $\Lambda^d\mathbb{C}^n$. This space is zero for $d > n$. Exercise 18.2 gives the exterior case in detail.

Theorem 18.2 (Schur–Weyl duality and the Schur character · *C*)**.** *Let $B \subseteq \operatorname{End}(T^d)$ be the span of all $g^{\otimes d}$ with $g \in GL_n(\mathbb{C})$.*

(a) $B = \operatorname{End}_{S_d}(T^d)$.

(b) Evaluation $\varphi \otimes v \mapsto \varphi(v)$ is an isomorphism

$$\bigoplus_{\lambda \vdash d} \mathbb{S}_\lambda(\mathbb{C}^n) \otimes V^\lambda \longrightarrow T^d$$

compatible with B and with S_d.

(c) $\mathbb{S}_\lambda(\mathbb{C}^n) \neq 0$ exactly when $\ell(\lambda) \leq n$. The nonzero ones are irreducible, pairwise nonisomorphic B-modules.

(d) For $\ell(\lambda) \leq n$, the trace of $\operatorname{diag}(x_1, \dots, x_n)^{\otimes d}$ on $\mathbb{S}_\lambda(\mathbb{C}^n)$ is the Schur polynomial s_λ, where

$$s_\lambda \, a_\delta = a_{\lambda+\delta}, \qquad a_\mu = \det\bigl(x_i^{\mu_j}\bigr)_{i,j=1}^{n}, \qquad \delta = (n-1, \dots, 1, 0).$$

Sources: the finite-group course, Lectures 13–14; Fulton–Harris, Section 6.1, Theorem 6.3, with proofs in Section 6.2; Goodman–Wallach, Section 9.1.

Weights. Let $\varepsilon_j(\operatorname{diag}(t)) = t_j$. The *content* μ of e_I counts indices: $\mu_j = \#\{k : i_k = j\}$. Then

$$\rho_d(\operatorname{diag}(t))\, e_I = \Bigl(\sum_j \mu_j t_j\Bigr) e_I, \qquad \operatorname{diag}(x)^{\otimes d} e_I = x^\mu e_I.$$

So the weight spaces $(T^d)_\mu$ of the Lie algebra agree with the joint eigenspaces of the operators $\operatorname{diag}(x)^{\otimes d}$ as x varies. Permutations preserve content, so each $(T^d)_\mu$ is S_d-stable and

$$\mathbb{S}_\lambda(\mathbb{C}^n)_\mu = \operatorname{Hom}_{S_d}\bigl(V^\lambda, (T^d)_\mu\bigr).$$

Taking traces in Theorem 18.2(d) gives

$$s_\lambda(x) = \sum_\mu \dim \mathbb{S}_\lambda(\mathbb{C}^n)_\mu \; x^\mu. \tag{18.1}$$

The Schur polynomial is the formal $\mathfrak{gl}_n$-character, written with monomials.

A finite direct sum of Schur modules $\mathbb{S}_\lambda(\mathbb{C}^n)$ is called a *polynomial* $\mathfrak{gl}_n$-module. The name fits because its character is a polynomial in $x_1, \dots, x_n$. Its exponents are contents, so they are nonnegative integers.

18.2 Irreducibility over $\mathfrak{gl}_n$ and $\mathfrak{sl}_n$

Irreducibility in Theorem 18.2(c) concerns the group span B. To use it for $\mathfrak{gl}_n$, we show that the operators $\rho_d(X)$ generate the algebra B.

Lemma 18.3 (Newton generation · *P*). *Let A be the unital associative subalgebra of $\mathrm{End}(T^d)$ generated by $\rho_d(\mathfrak{gl}_n)$. Then $A = \mathrm{End}_{S_d}(T^d)$.*

Proof. Every $\rho_d(X)$ commutes with S_d, so $A \subseteq \mathrm{End}_{S_d}(T^d)$.

Fix $X \in \mathfrak{gl}_n$. The operators $X_{(1)}, \dots, X_{(d)}$ commute pairwise. Let p_m and e_m be their power sums and elementary symmetric polynomials. Since $X_{(k)}^m = (X^m)_{(k)}$,

$$p_m = \sum_{k=1}^{d} (X^m)_{(k)} = \rho_d(X^m) \in A.$$

Newton's identities hold for commuting variables over $\mathbb{Q}$:

$$m\, e_m = \sum_{j=1}^{m} (-1)^{j-1} e_{m-j}\, p_j, \qquad e_0 = 1.$$

Division by m is allowed over $\mathbb{C}$. The first two cases read

$$e_1 = p_1, \qquad e_2 = \frac{1}{2}\bigl(e_1 p_1 - e_0 p_2\bigr) = \frac{1}{2}\bigl(p_1^2 - p_2\bigr).$$

In general, e_m is a polynomial in $e_0, \dots, e_{m-1}$ and $p_1, \dots, p_m$. Induction on m gives $e_m \in A$. In particular

$$e_d = X_{(1)} X_{(2)} \cdots X_{(d)} = X^{\otimes d} \in A.$$

Taking $X = g \in GL_n(\mathbb{C})$ gives $B \subseteq A$. By Theorem 18.2(a), $\mathrm{End}_{S_d}(T^d) = B \subseteq A$. ∎

Theorem 18.4 (Irreducibility · *P*). *Let $\lambda \vdash d$ with $\ell(\lambda) \le n$. Then $\mathbb{S}_\lambda(\mathbb{C}^n)$ is an irreducible $\mathfrak{gl}_n$-module, and the identity matrix I acts by the scalar d. Its restriction to $\mathfrak{sl}_n$ is irreducible.*

Proof. A subspace of $\mathbb{S}_\lambda(\mathbb{C}^n)$ is $\mathfrak{gl}_n$-stable exactly when it is stable under post-composition by A. By Lemma 18.3 and Theorem 18.2(a), $A = B$. By Theorem 18.2(c), $\mathbb{S}_\lambda(\mathbb{C}^n)$ is an irreducible B-module. Hence it is an irreducible $\mathfrak{gl}_n$-module.

The identity acts on T^d by

$$\rho_d(I) = \sum_{k=1}^{d} I_{(k)} = d\ \mathrm{id}\,.$$

Finally, every $X \in \mathfrak{gl}_n$ splits as

$$X = \Bigl(X - \frac{\operatorname{tr} X}{n} I\Bigr) + \frac{\operatorname{tr} X}{n} I, \qquad X - \frac{\operatorname{tr} X}{n} I \in \mathfrak{sl}_n.$$

Hence $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$. The second summand acts by scalars, which preserve every subspace. So every $\mathfrak{sl}_n$-stable subspace is $\mathfrak{gl}_n$-stable, and irreducibility passes to $\mathfrak{sl}_n$. ∎

Remark 18.5. The proof uses the scalar action of I. General $\mathfrak{gl}_n$-modules need not be completely reducible. Exercise 18.5 examines the role of the centre.

18.3 Highest weights and the Weyl character formula

Let $\mathfrak{h} \subseteq \mathfrak{sl}_n$ be the traceless diagonal matrices and $\bar{\varepsilon}_j = \varepsilon_j|_{\mathfrak{h}}$. Then $\sum_j \bar{\varepsilon}_j = 0$, so $\bar{\mu} = \sum_j \mu_j \bar{\varepsilon}_j$ depends on μ only modulo constant vectors. The simple roots and coroots are

$$\alpha_i = \bar{\varepsilon}_i - \bar{\varepsilon}_{i+1}, \qquad h_i = E_{ii} - E_{i+1,i+1}, \qquad 1 \le i \le n-1.$$

Since $\langle \bar{\varepsilon}_j, h_i \rangle = \delta_{ij} - \delta_{j,i+1}$,

$$\langle \bar{\mu}, h_i \rangle = \sum_{j=1}^{n} \mu_j \big(\delta_{ij} - \delta_{j,i+1}\big) = \mu_i - \mu_{i+1}.$$

Hence the fundamental weight dual to h_i is $\omega_i = \bar{\varepsilon}_1 + \cdots + \bar{\varepsilon}_i$. For $\mu \in \mathbb{Z}^n$, the weight $\bar{\mu}$ is dominant integral exactly when $\mu_1 \ge \cdots \ge \mu_n$.

The Weyl vector ρ is half the sum of the positive roots $\bar{\varepsilon}_i - \bar{\varepsilon}_j$ $(i < j)$. We find it by counting. The weight $\bar{\varepsilon}_j$ occurs with sign $+$ in the $n - j$ roots with first index j. It occurs with sign $-$ in the $j - 1$ roots with second index j. Hence

$$\begin{aligned} \rho &= \frac{1}{2} \sum_{j=1}^{n} \big((n-j) - (j-1)\big) \bar{\varepsilon}_j \\ &= \sum_{j=1}^{n} \frac{n+1-2j}{2}\, \bar{\varepsilon}_j \\ &= \sum_{j=1}^{n} (n-j)\, \bar{\varepsilon}_j = \bar{\delta}. \end{aligned}$$

The last step adds $\frac{n-1}{2} \sum_j \bar{\varepsilon}_j = 0$. The Weyl group S_n acts by $w\bar{\varepsilon}_j = \bar{\varepsilon}_{w(j)}$, with $\det(w) = \operatorname{sgn}(w)$.

Define the ring homomorphism

$$\pi : \mathbb{Z}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}] \longrightarrow \mathbb{Z}[P], \qquad x^{\mu} \longmapsto e^{\bar{\mu}}.$$

Expanding the determinant a_μ over permutations gives

$$a_\mu = \sum_{w \in S_n} \operatorname{sgn}(w) \prod_{j=1}^{n} x_{w(j)}^{\mu_j}.$$

The monomial for w has exponent μ_j at the variable $x_{w(j)}$. Its image under π is

$$\pi\Big(\prod_j x_{w(j)}^{\mu_j}\Big) = e^{\sum_j \mu_j \bar{\varepsilon}_{w(j)}} = e^{w\bar{\mu}}.$$

Summing with signs,

$$\pi(a_\mu) = \sum_{w \in S_n} \det(w)\, e^{w\bar{\mu}} = A_{\bar{\mu}}.$$

The map π restricts each weight from the diagonal matrices to $\mathfrak{h}$. Hence, by (18.1), $\pi(s_\lambda)$ is the $\mathfrak{sl}_n$-character of $\mathbb{S}_\lambda(\mathbb{C}^n)$. Apply the ring homomorphism π to $s_\lambda a_\delta = a_{\lambda+\delta}$:

$$\begin{aligned} \pi(s_\lambda)\, \pi(a_\delta) &= \pi(a_{\lambda+\delta}), \\ \operatorname{ch} \mathbb{S}_\lambda(\mathbb{C}^n) \cdot A_{\bar{\delta}} &= A_{\bar{\lambda}+\bar{\delta}}. \end{aligned}$$

Since $\bar{\delta} = \rho$, this is

$$\operatorname{ch} \mathbb{S}_\lambda(\mathbb{C}^n) \cdot A_\rho = A_{\bar{\lambda}+\rho}. \tag{18.2}$$

The Weyl character formula of Lecture 16 reads $\operatorname{ch} L(\nu) \cdot A_\rho = A_{\nu+\rho}$. The dictionary is

$$x^{\mu} \leftrightarrow e^{\bar{\mu}}, \qquad \delta \leftrightarrow \rho, \qquad a_\mu \leftrightarrow A_{\bar{\mu}}, \qquad S_n \leftrightarrow W.$$

Lemma 18.6 (Leading term · *P*). *Let $l_1 > \cdots > l_n$ be integers and $\gamma = \sum_j l_j \bar{\varepsilon}_j$. The weights $w\gamma$, $w \in S_n$, are distinct, and only $w = 1$ gives a dominant weight.*

Proof. Reindex the sum by $k = w(j)$:

$$w\gamma = \sum_{j=1}^{n} l_j \bar{\varepsilon}_{w(j)} = \sum_{k=1}^{n} l_{w^{-1}(k)} \bar{\varepsilon}_k.$$

So $w\gamma$ has coordinate sequence $(l_{w^{-1}(1)}, \ldots, l_{w^{-1}(n)})$ modulo constants. Suppose two such sequences differ by a constant c in each entry. Both have entry sum $l_1 + \cdots + l_n$, so $nc = 0$ and $c = 0$. Since the l_j are distinct, equal sequences come from equal permutations. A sequence of distinct integers is weakly decreasing only in the order $(l_1, \ldots, l_n)$, that is, for $w = 1$. ∎

Theorem 18.7 (Highest weight · *P*). *Let $\lambda \vdash d$ with $\ell(\lambda) \le n$. As $\mathfrak{sl}_n$-modules,*

$$\mathbb{S}_\lambda(\mathbb{C}^n) \cong L(\bar{\lambda}), \qquad \bar{\lambda} = \sum_{i=1}^{n-1} (\lambda_i - \lambda_{i+1})\, \omega_i.$$

As a $\mathfrak{gl}_n$-module it has highest weight $\sum_j \lambda_j \varepsilon_j$.

Proof. By Theorem 18.4 and the classification of Lecture 13, $\mathbb{S}_\lambda(\mathbb{C}^n) \cong L(\nu)$ for some dominant integral ν. The Weyl character formula and (18.2) give

$$A_{\nu+\rho} = \operatorname{ch} L(\nu) \cdot A_\rho = \operatorname{ch} \mathbb{S}_\lambda(\mathbb{C}^n) \cdot A_\rho = A_{\bar{\lambda}+\rho}.$$

Write $\nu = \sum_i m_i \omega_i$ with $m_i \ge 0$. Since $\omega_i = \bar{\varepsilon}_1 + \cdots + \bar{\varepsilon}_i$ and $\rho = \sum_j (n-j)\bar{\varepsilon}_j$,

$$\nu + \rho = \sum_{j=1}^{n} l_j \bar{\varepsilon}_j, \qquad l_j = \sum_{i=j}^{n-1} m_i + (n-j),$$

with an empty sum for $j = n$. Then $l_j - l_{j+1} = m_j + 1 \ge 1$, so the coordinates strictly decrease. The same holds for $\bar{\lambda} + \rho$. Here $\bar{\lambda} = \sum_j \lambda_j \bar{\varepsilon}_j$, whose labels are $\lambda_i - \lambda_{i+1}$ by the pairing with h_i computed above. So $\bar{\lambda} + \rho$ has coordinates $\lambda_j + n - j$. By Lemma 18.6, the only dominant exponential on the left is $e^{\nu+\rho}$, and on the right it is $e^{\bar{\lambda}+\rho}$. Hence $\nu = \bar{\lambda}$.

A highest vector v is killed by the strictly upper triangular matrices, which lie in both $\mathfrak{sl}_n$ and $\mathfrak{gl}_n$. Since I acts by a scalar, v is also an eigenvector of every diagonal matrix in $\mathfrak{gl}_n$. Its $\mathfrak{gl}_n$-weight μ restricts to $\bar{\lambda}$, so $\mu = \lambda + c(1, \ldots, 1)$ for some $c \in \mathbb{C}$. The identity acts by $d = |\lambda|$, so

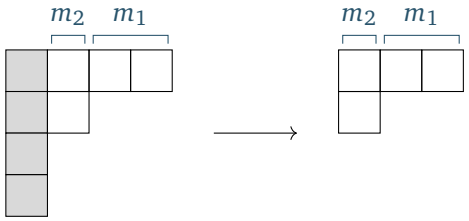
$$|\lambda| = \mu(I) = \sum_{j=1}^{n} \mu_j = \sum_{j=1}^{n} \lambda_j + nc = |\lambda| + nc.$$

Hence $c = 0$ and $\mu = \lambda$. ∎

So a partition represents an $\mathfrak{sl}_n$ highest weight, and its Dynkin labels are the differences of consecutive rows. Diagrammatically, $m_i = \lambda_i - \lambda_{i+1}$ is the number of columns of height exactly i.

18.4 Determinant twists and normalised partitions

The partition (1^n) is a single column of height n, called a *full column*. Figure 18.1 removes a full column for $n = 4$ and counts the remaining columns by height.



$\lambda = (4,2,1,1)$ $\quad$ $\lambda = (3,1,0,0)$
$(m_1, m_2, m_3) = (2,1,0)$ $\quad$ $(m_1, m_2, m_3) = (2,1,0)$

Figure 18.1: Removing the shaded full column from $(4,2,1,1)$ leaves $(3,1,0,0)$. A bracket labelled m_i marks the columns of height exactly i, and no column has height 3, so both diagrams have the Dynkin labels $(2,1,0)$.

Adding 1 to every exponent multiplies row i of the matrix $(x_i^{\mu_j})$ by x_i. Factoring x_i from row i gives

$$a_{\lambda+(1^n)+\delta} = \det\big(x_i \cdot x_i^{\lambda_j+n-j}\big) = x_1 \cdots x_n \, a_{\lambda+\delta}.$$

Dividing by a_δ, and taking $\lambda = \varnothing$ for the second formula, gives

$$s_{\lambda+(1^n)} = x_1 \cdots x_n \, s_\lambda, \qquad s_{(1^n)} = x_1 \cdots x_n.$$

Thus $\mathbb{S}_{(1^n)}(\mathbb{C}^n)$ is one-dimensional. Its Dynkin labels are all zero, so Theorem 18.7 makes it the trivial $\mathfrak{sl}_n$-module. By Theorem 18.4, the identity acts by n. Split X as in that proof. The traceless part acts by 0, since the module is trivial over $\mathfrak{sl}_n$. Hence

$$X = \Big(X - \frac{\operatorname{tr} X}{n} I\Big) + \frac{\operatorname{tr} X}{n} I \quad \text{acts by} \quad 0 + \frac{\operatorname{tr} X}{n} \cdot n = \operatorname{tr} X.$$

The top exterior power is the same module. The matrix X acts on $e_1 \wedge \cdots \wedge e_n$ by replacing one factor e_j at a time by $Xe_j = \sum_i X_{ij} e_i$. A term with $i \neq j$ repeats e_i and vanishes, so

$$X \cdot (e_1 \wedge \cdots \wedge e_n) = \Big(\sum_{j=1}^{n} X_{jj}\Big) e_1 \wedge \cdots \wedge e_n.$$

Hence $\mathbb{S}_{(1^n)}(\mathbb{C}^n) \cong \Lambda^n \mathbb{C}^n$. We call this module det.

Proposition 18.8 (Full columns · *P*)**.** *Let λ, μ be partitions of length at most n.*

(i) $\mathbb{S}_\lambda(\mathbb{C}^n) \cong \mathbb{S}_\mu(\mathbb{C}^n)$ *as $\mathfrak{sl}_n$-modules exactly when $\lambda - \mu$ is a constant vector.*

(ii) They are isomorphic as $\mathfrak{gl}_n$-modules exactly when $\lambda = \mu$.

(iii) $\mathbb{S}_{\lambda+(1^n)}(\mathbb{C}^n) \cong \mathbb{S}_\lambda(\mathbb{C}^n) \otimes \det$ *as $\mathfrak{gl}_n$-modules.*

Proof. (i) By Theorem 18.7 and uniqueness in Lecture 13, the modules are isomorphic exactly when $\bar\lambda = \bar\mu$. This means $\lambda_i - \lambda_{i+1} = \mu_i - \mu_{i+1}$ for all i, that is, when $\lambda - \mu$ is a constant vector.

(ii) A $\mathfrak{gl}_n$-isomorphism is an $\mathfrak{sl}_n$-isomorphism, so $\lambda - \mu = c(1^n)$ by (i). Summing the entries gives the first line below. The isomorphism also matches the scalars by which I acts, which gives the second:

$$|\lambda| = |\mu| + nc,$$
$$|\lambda| = |\mu|.$$

Hence $nc = 0$ and $\lambda = \mu$. The converse is clear.

(iii) Both sides restrict to $L(\bar{\lambda})$ on $\mathfrak{sl}_n$, since adding a full column keeps the labels and det is trivial on $\mathfrak{sl}_n$. The identity I acts on both by $|\lambda| + n$. An $\mathfrak{sl}_n$-isomorphism commutes with scalars, so it commutes with $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$. ∎

The *length cutoff* is the condition $\ell(\lambda) \le n$ in Theorem 18.2(c). In other words, no column is longer than n. By Proposition 18.8, a full column of height n does not change the $\mathfrak{sl}_n$-module. Over $\mathfrak{gl}_n$ it changes the module by a determinant twist. In Figure 18.1, $\mathbb{S}_{(4,2,1,1)}(\mathbb{C}^4) \cong \mathbb{S}_{(3,1)}(\mathbb{C}^4) \otimes \det$. A partition is *normalised* if $\lambda_n = 0$, that is, $\ell(\lambda) \le n - 1$.

Theorem 18.9 (All simple $\mathfrak{sl}_n$-modules · *P*). *The rule $\lambda \mapsto \mathbb{S}_\lambda(\mathbb{C}^n)$ is a bijection from normalised partitions to isomorphism classes of simple $\mathfrak{sl}_n$-modules. The inverse sends $L(\sum_i m_i\omega_i)$ to*

$$\lambda_j = m_j + m_{j+1} + \cdots + m_{n-1} \quad (1 \le j \le n-1), \qquad \lambda_n = 0.$$

Proof. For $m_i \ge 0$, the displayed λ is weakly decreasing, has $\lambda_n = 0$, and satisfies

$$\lambda_i - \lambda_{i+1} = (m_i + \cdots + m_{n-1}) - (m_{i+1} + \cdots + m_{n-1}) = m_i.$$

So λ is a normalised partition with labels m_i. Conversely, a normalised partition is recovered from its differences, since $\lambda_n = 0$. All $m_i = 0$ gives $\varnothing$ and the trivial module. Theorem 18.7 and the bijection of Lecture 13 complete the proof. ∎

Every partition of length at most n becomes normalised after removing λ_n full columns. For $\mathfrak{gl}_n$, twisting by $\det^k$ with $k \in \mathbb{Z}$, where $\det^{-1} = (\Lambda^n\mathbb{C}^n)^*$, gives highest weights $\lambda_1 \ge \cdots \ge \lambda_n$ in $\mathbb{Z}$. One-dimensional modules $X \mapsto s\,\mathrm{tr}\,X$ with $s \notin \mathbb{Z}$ and the nonsemisimple example of Exercise 18.5 are further $\mathfrak{gl}_n$-modules outside this family.

18.5 Young symmetrisers and the adjoint module of $\mathfrak{sl}_3$

The Schur module is defined as a space of homomorphisms, so its elements are maps and not tensors. Explicit calculations need actual tensors. A Young symmetriser supplies them. It is an element of $\mathbb{C}[S_d]$, and we show that its image in T^d is a copy of the Schur module. Lecture 20 uses this copy to treat $\mathbb{S}_\lambda(\mathbb{C}^n)$ as a submodule of T^d, and Lecture 22 imposes further linear conditions on these tensors.

Let T_λ be the tableau of shape λ numbered $1, \dots, d$ along rows. Let R and C be its row and column groups, and set

$$a = \sum_{p \in R} p, \qquad b = \sum_{q \in C} \operatorname{sgn}(q)\, q, \qquad c_\lambda = ab \in \mathbb{C}[S_d].$$

Theorem 18.10 (Young symmetriser · *C*). *For every $\lambda \vdash d$, $c_\lambda^2 = k_\lambda c_\lambda$ with $k_\lambda = d!/\dim V^\lambda$, and the left ideal $\mathbb{C}[S_d]c_\lambda$ is isomorphic to V^λ. Source: Fulton–Harris, Section 4.1, Theorem 4.3, recalled on the Young-symmetriser sheet of the finite-group course.*

Proposition 18.11 (Image model · *P*). *$\mathbb{S}_\lambda(\mathbb{C}^n) \cong c_\lambda T^d$ as $\mathfrak{gl}_n$-modules.*

Proof. Put $\epsilon = c_\lambda/k_\lambda$, so $\epsilon^2 = \epsilon$ and $\mathbb{C}[S_d]\epsilon \cong V^\lambda$. Define

$$\Phi\colon \operatorname{Hom}_{S_d}(\mathbb{C}[S_d]\epsilon, T^d) \longrightarrow \epsilon T^d, \qquad \Phi(\varphi) = \varphi(\epsilon) = \epsilon\varphi(\epsilon).$$

The second equality holds because φ is S_d-linear and $\epsilon^2 = \epsilon$, so $\Phi(\varphi) \in \epsilon T^d$. For $t \in \epsilon T^d$ define $\psi_t \colon \mathbb{C}[S_d]\epsilon \to T^d$ by $\psi_t(y\epsilon) = yt$ for $y \in \mathbb{C}[S_d]$. This map is well defined. If $y\epsilon = 0$, then $yt = y\epsilon t = 0$, since $t = \epsilon t$. It is S_d-linear, and the two constructions are inverse. For $t \in \epsilon T^d$ and $\varphi \in \operatorname{Hom}_{S_d}(\mathbb{C}[S_d]\epsilon, T^d)$,

$$\Phi(\psi_t) = \psi_t(\epsilon) = \epsilon t = t,$$
$$\psi_{\varphi(\epsilon)}(y\epsilon) = y\varphi(\epsilon) = \varphi(y\epsilon).$$

The last equality uses the S_d-linearity of φ.

Since $\rho_d(X)$ commutes with S_d, it preserves ϵT^d, and

$$\Phi\big(\rho_d(X) \circ \varphi\big) = \rho_d(X)\varphi(\epsilon) = \rho_d(X)\Phi(\varphi).$$

So Φ is $\mathfrak{gl}_n$-linear. Composing with an isomorphism $V^\lambda \cong \mathbb{C}[S_d]\epsilon$ identifies the domain of Φ with $\mathbb{S}_\lambda(\mathbb{C}^n)$. Finally $\epsilon T^d = c_\lambda T^d$, since $k_\lambda \neq 0$. ∎

Example 18.12 ($\lambda = (2,1)$, $n = 3$). *The character.* Here $\lambda + \delta = (4,2,0)$. With $a_\delta = \prod_{i<j}(x_i - x_j)$, the alternant $a_{(4,2,0)}$ is the Vandermonde determinant in the variables x_i^2:

$$\begin{aligned} a_{(4,2,0)} &= \prod_{i<j}(x_i^2 - x_j^2) \\ &= \prod_{i<j}(x_i - x_j)\prod_{i<j}(x_i + x_j) \\ &= a_{(2,1,0)}\prod_{i<j}(x_i + x_j). \end{aligned}$$

Dividing by $a_{(2,1,0)}$ and expanding,

$$\begin{aligned} s_{(2,1)} &= (x_1 + x_2)(x_1 + x_3)(x_2 + x_3) \\ &= (x_1^2 + x_1x_2 + x_1x_3 + x_2x_3)(x_2 + x_3) \\ &= \sum_{i \neq j} x_i^2 x_j + 2x_1x_2x_3. \end{aligned}$$

If $\{i,j,k\} = \{1,2,3\}$, then $\bar\varepsilon_i + \bar\varepsilon_j + \bar\varepsilon_k = 0$ gives

$$\pi(x_i^2 x_j) = e^{2\bar\varepsilon_i + \bar\varepsilon_j} = e^{\bar\varepsilon_i - \bar\varepsilon_k}.$$

As (i,k) runs over ordered pairs of distinct indices, the six monomials give the six roots, and $x_1x_2x_3$ gives weight 0 twice. This is the adjoint character of Lecture 16. The Dynkin labels are $(2-1, 1-0) = (1,1)$. The Weyl dimension formula of Lecture 17 with $p = q = 1$ gives

$$\dim L(\omega_1 + \omega_2) = \frac{(1+1)(1+1)(1+1+2)}{2} = 8.$$

A highest vector. Write $e_{ijk} = e_i \otimes e_j \otimes e_k$. For $T_\lambda = \begin{smallmatrix} 1 & 2 \\ 3 \end{smallmatrix}$,

$$c_\lambda = \big(1 + (12)\big)\big(1 - (13)\big).$$

The transpositions (13) and (12) swap the first factor with the third and with the second:

$$\begin{aligned} \big(1 - (13)\big)e_{112} &= e_{112} - e_{211}, \\ u = c_\lambda e_{112} &= \big(1 + (12)\big)\big(e_{112} - e_{211}\big) \\ &= e_{112} - e_{211} + e_{112} - e_{121} \\ &= 2e_{112} - e_{211} - e_{121}. \end{aligned}$$

The operator $\rho_3(E_{12})$ replaces one factor e_2 by e_1, and $\rho_3(E_{23})$ kills vectors without e_3:

$$\begin{aligned}\rho_3(E_{12})u &= 2\rho_3(E_{12})e_{112} - \rho_3(E_{12})e_{211} - \rho_3(E_{12})e_{121}\\ &= 2e_{111} - e_{111} - e_{111} = 0,\\ \rho_3(E_{23})u &= 0.\end{aligned}$$

Since $E_{13} = [E_{12}, E_{23}]$, the vector u is killed by all of $\mathfrak{n}_+$. The coefficient of e_{112} in u is 2, so $u \neq 0$. Thus $u \in c_\lambda T^3$ is a highest vector of weight $2\varepsilon_1 + \varepsilon_2$.

Exercise 18.4(v) compares the S_3-factor with the zero weight space of the Schur factor.

Self-checks

(1) Which $\mathfrak{sl}_4$-module is $\mathbb{S}_{(3,3,1,1)}(\mathbb{C}^4)$?

(2) Why is $\mathbb{S}_{(1,1,1,1)}(\mathbb{C}^3) = 0$, while $\mathbb{S}_{(2,1,1)}(\mathbb{C}^3) \cong \mathbb{C}^3$ for $\mathfrak{sl}_3$?

(3) How many independent highest vectors of weight $2\varepsilon_1 + \varepsilon_2$ lie in $(\mathbb{C}^3)^{\otimes 3}$?

Answers. (1) The labels are $(0, 2, 0)$, so it is $L(2\omega_2)$. The normalised partition is $(2, 2)$, and over $\mathfrak{gl}_4$ it is $\mathbb{S}_{(2,2)} \otimes \det$. (2) The first has length $4 > 3$. The second has labels $(1, 0)$; removing its full column leaves (1). (3) Two. They are $ae_{112} + be_{121} + ce_{211}$ with $a + b + c = 0$, matching $\dim V^{(2,1)} = 2$.

Summary

Newton's identities show that the operators $\rho_d(X)$ generate the commutant $\mathrm{End}_{S_d}(T^d)$. Hence, for $\ell(\lambda) \leq n$, the module $\mathbb{S}_\lambda(\mathbb{C}^n)$ is irreducible over $\mathfrak{gl}_n$. It stays irreducible over $\mathfrak{sl}_n$ because I acts by the scalar $|\lambda|$.

The bialternant is the Weyl character formula in coordinates. It identifies $\mathbb{S}_\lambda(\mathbb{C}^n)$ with $L(\bar{\lambda})$, with labels $\lambda_i - \lambda_{i+1}$. Normalised partitions give every simple $\mathfrak{sl}_n$-module. Adding a full column tensors the $\mathfrak{gl}_n$-module with det.

Reading

The finite-group course, Lectures 13–14. Fulton and Harris, *Representation Theory: A First Course*, Section 4.1, Sections 6.1–6.2 and Sections 15.2–15.3. Goodman and Wallach, *Symmetry, Representations, and Invariants*, Sections 9.1 and 9.3.